Huiqin (Hermine) Ni

Deep-Tech R&D | Polymer Materials | Functional Interfaces

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LinkedIn | Google Scholar | GitHub

PROFESSIONAL SUMMARY

Mechanical Engineering Ph.D. researcher with training across materials science, chemical engineering, and applied physics. Research experience spans responsive polymers, functional coatings, wetting, adhesion, soft actuators, and nanomaterials. Brings hands-on materials fabrication and characterization, experimental design, data analysis, peer-reviewed publication, and international research collaboration.

CORE EXPERTISE

Materials: Responsive polymers, functional surfaces, slippery coatings, bioinspired adhesion, hydrogels, MXenes, and nanocomposites

R&D capabilities: Materials preparation, experimental design, surface and mechanical characterization, comparative testing, data interpretation, and technical communication

EDUCATION

North Carolina State University **Aug 2024 – Present**
Ph.D. in Mechanical Engineering | Raleigh, NC

China University of Petroleum **Aug 2021 – Jun 2024**
M.S. in Chemical Engineering and Technology | Beijing, China

Aalto University **Sep 2022 – Aug 2023**
Visiting Student, Applied Physics | Helsinki, Finland

China University of Petroleum **Aug 2017 – Jun 2021**
B.Eng. in Materials Science and Engineering | Rank: 2/84

SELECTED RESEARCH EXPERIENCE

Ph.D. Research - Functional Polymer Interfaces **Aug 2024 – Present**
North Carolina State University

- Conduct experimental research on functional polymer surfaces and interfacial behavior using materials preparation, experimental design, surface characterization, and data analysis.
- Communicate technical findings through reports, presentations, and cross-disciplinary collaboration; project-level details are shared selectively until publication.

EARLIER RESEARCH EXPERIENCE

Multi-Mode Shape-Memory Polymer Actuators **Oct 2022 – Jun 2024**
China University of Petroleum / Aalto University

- Investigated responsive shape-memory polymers for soft actuation through fabrication, characterization, mechanical testing, and technical writing.

MXene Processing for Printed and Wearable Electronics **Oct 2022 – Jun 2023**
Aalto University

- Compared solvent-exchange conditions for MXene processing toward printed and wearable electronics.

Controllable Bioinspired Adhesion for Robotic Grippers **Dec 2020 – Jan 2022**

- Developed gecko-inspired PDMS surfaces; fabricated and characterized samples and contributed to technical manuscript drafting.

Responsive Hydrogels and Photocatalytic Materials **Apr 2019 – May 2022**

- Investigated multi-field-responsive PNIPAM nanocomposite hydrogels and defect-engineered BiOBr photocatalysts.

SELECTED PUBLICATIONS

- Rather, A. M., et al., including **H. Ni**. "Liquid crystal elastomer-based reversible metamorphosis for loss-less droplet manipulation." *Materials Horizons* 13(6), 2946–2957 (2026).
- Mi, Y., et al., including **H. Ni**. "Gecko-inspired reversible adhesion via quantum dots enabled photo-detachment." *Chemical Engineering Journal* 431, 134081 (2022).
- Luo, X., et al., including **H. Ni**. "Near-infrared responsive gecko-inspired flexible arm gripper." *Materials Today Physics* 29, 100919 (2022).
- Four peer-reviewed journal articles and two book contributions; full publication record available on Google Scholar.

TECHNICAL SKILLS

Fabrication: Polymer synthesis and processing, hydrogel fabrication, nanomaterials, plasma treatment, surface modification, laser processing, coatings, and 3D printing

Characterization: Contact and sliding-angle goniometry, FT-IR, AFM, SEM, UV-Vis, XRD, DSC, TGA, DMA, and Instron testing

Computational and engineering tools: Python, MATLAB, C, Linux/Ubuntu, Git/GitHub, ns-3, SolidWorks, CAD, Abaqus, Origin, and LaTeX

Languages: Chinese (native), English (professional working proficiency), and Japanese (basic)

SELECTED RECOGNITION

Leon Memorial Scholarship, NC State College of Engineering (2026); Materials 4.0 Summer School Scholarship (2023); Outstanding Graduate (2021).